

The Three Questions

Immanuel Kant, *Critique of Pure Reason* (1781)

Carlos Cotrini

A land of truth in a sea of illusion



C. D. Friedrich,

Wanderer above the Sea of Fog (c. 1817)

Es ist das **Land der Wahrheit** (ein reizender Name), umgeben von einem weiten und stürmischen Oceane, dem eigentlichen Sitze des Scheins, wo manche Nebelbank und manches bald wegschmelzende Eis neue Länder lügt ...

It is the *land of truth* (an enchanting name), surrounded by a wide and stormy ocean, the very seat of illusion, where many a fog bank and many a swiftly melting iceberg feign new lands ...

Kritik der reinen Vernunft, A235 / B294 (1781 / 1787)

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Nebelbank

Weekend 1: honest uncertainty

wegschmelzendes Eis

Weekend 2: ground that is not there

[Land der Wahrheit](#)

[Weekends 5 to 7: today](#)

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Weekend 1 was the fog. Weekend 2 was the ice. Now we go ashore.

What the five weekends behind us did

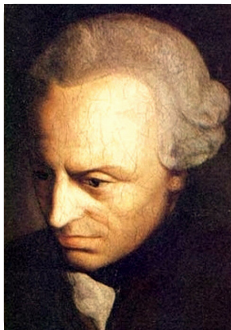
Weekend 0	NumPy, scikit-learn, PyTorch	the tools
Weekend 1	entropy, KL, variance, ensembles	how to measure what you do not know
Weekend 2	confounding, selection, attribution	why a model that scores well can be wrong
Weekend 3	attacks, jailbreaks, privacy	that somebody is trying to break it
Weekend 4	policy gradients, actor-critic, PPO	how a machine chooses under uncertainty

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None of them let it look anything up, do anything, or run at a size anyone would pay for.

All the interest of my reason



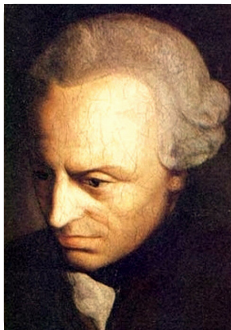
Immanuel Kant (c. 1790),
painter unknown

Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische) vereinigt sich in folgenden drei Fragen:

- 1.
- 2.
- 3.

Kritik der reinen Vernunft, A805 / B833; translation C. Cotrini

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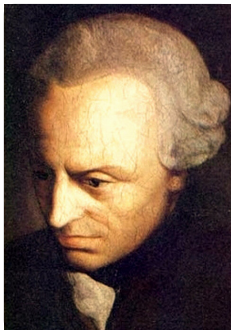
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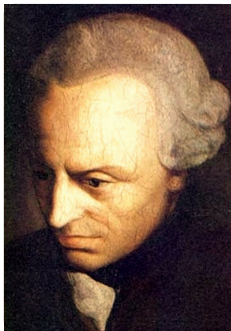
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The three questions, and the three weekends

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Was kann ich wissen? → Weekend 5: RAG

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Kant glosses the third himself: *wenn ich nun tue, was ich soll, was darf ich alsdann hoffen?*

If I now do what I ought, what may I then hope? That is the shape of a scaling law.

Why these three, and why now

- **Weekend 5.** A trained model knows only what was in its weights the day training stopped. Your company, your customers, your last quarter: all outside the weights. Retrieval is what puts them inside the answer.

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Weekends 1 to 4 were about a model. Weekends 5 to 7 are about a system.

Kant printed the list again, with one more question

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|-------------------------|------------|
| 1. Was kann ich wissen? | Metaphysik |
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Im Grunde könnte man aber alles dieses zur Anthropologie rechnen, weil die drei ersten Fragen sich auf die letzte beziehen.

At bottom one could reckon all of this to anthropology, because the first three questions relate to the last.

Jäsche *Logik* (1800), Einleitung III, AA IX:25; translation C. Cotrini

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We will answer the first three. Keep the fourth in view.